YREC DOCUMENTATION

1. **What YREC is**. YREC is a Fortran stellar evolution code optimized for the study of hydrostatic stars from the pre-main sequence through to the helium burning phase of evolution. YREC can also be used to study giant planets and white dwarfs. It is fast, precise, and accurate in the domains where it functions. YREC has been extensively used for both stellar population and stellar physics studies, including precision solar models. It includes important physical processes not traditionally included in classical stellar models (see below).
   Physical processes: YREC uses modern input microphysics (equation of state, energy transport, energy generation, surface boundary conditions) with a variety of user-configurable options. YREC is a pseudo-2D code that includes gravitational settling, mass loss, rotation, angular momentum loss from magnetized winds, and the structural effects of magnetic fields.
2. **Read @@YREC start guide for installation and run instructions**.
3. **Global Namelist Variable Types:**

- Variables starting with L are logical true/false variables
- Variables starting with I, J, K, M, N are integers
- Variables starting with other letters are floating point numbers, with exceptions noted below.
- Variables starting with F are sometimes character strings, typically pointers to input or output files.
- Variables starting with A are sometimes character strings.

Run Instructions and I/O controls: yrec8.nml1

YREC has two mandatory namelists. The second namelist, yrec8.nml2, discussed in detail below, contains almost all of the physical options (e.g. does the model include rotation? What are the assumed nuclear reaction cross-sections?) and the numerical ones (what are the adopted spatial and temporal resolutions? How accurate does the solution need to be?) The first namelist contains global run instructions, pointers to the input and output files, and specifies the frequency of output and its degree of detail. The first namelist also includes a small number of miscellaneous controls , described below.

Global Instructions

DESCRIP(1) – (2): A two-line plain text header describing the run. This appears as a header in the output track file.

Variables Specifying the Main Properties of the Run

YREC is a relaxation code that requires a valid solution of the structure equations as an input. There are therefore three classes of operations that it can perform:

1) It can rescale a starting model (as described below) with a zero timestep. This is appropriate for the zero-age main sequence (core H burning) or the zero-age horizontal branch (core He burning).
2) It can rescale and evolve a starting model with a non-zero timestep. This is required for pre-MS models, as there is no time-independent solution for a star supported by gravitational potential energy release.
3) It can evolve a starting model, or the result of a prior rescaling step, without changing the global model properties.

A typical run will involve 2 steps: a rescaling of the starting model to the correct mixing length, composition, mass, and input physics (which are in general different than the ones assumed in the library), followed by a run to a specified end point or number of models. If the parameters of the rescaled model are very different from the input, multiple rescalings may be needed (example: using a solar metallicity model as a template for a very metal poor model may need a series of metallicity rescalings).

Each YREC run has a fixed mass and composition (set in namelist 1), and a starting rotation rate (set in namelist 2). Suites of models with varying mass, composition, or rotation rate are generated by multiple runs of YREC using scripts.

Mandatory global variables are:

NUMRUN (n/a) - The number of separate runs the code will execute. For each of these runs, the following vectors *must* be specified:

KINDRN(I) – The type of operation being performed. 1 = evolve (all rescaling options are ignored in this case); 2 = rescale (rescaling options used, timestep set to zero); 3 =

evolve and rescale (rescaling options used, timestep calculated as normal). Option 3 should only be used for the pre-MS.

LFIRST(I) – This T/F flag specifies whether the starting model will be used, or if the result of the prior run will be used. This is hard-coded true on the first run. In general, this will be TRUE for the first run and false for subsequent ones. (One can do clever exceptions, but it is almost always better practice to use multiple code runs instead.)

NMODLS(I) – The code will run to this number of models, until it crashes, or if it reaches an optional independent stop condition.

CMIXLA(I) – The mixing length, in pressure scale heights. If fixed for a calculation, specify the same value in all runs.

ZENV0A(I) - The surface metallicity; this is used as an output label, not an input variable

XENV0A(I) - The surface hydrogen mass fraction; this is used as an output label, not an input variable.

Rescaling variables are:

A negative value means that this component is not being rescaled (except for SENV0A, which by definition is negative). **Note: care should be taken for CHeB stars.**

RSCLM(I) - If positive, rescale to this total mass (Msun). All mass variables are stretched in a pre-MS or MS model. For a CHeB star the envelope mass is stretched and the He core mass is held fixed (see below).

RSCLCM(I) - If positive, rescale to this He core mass (CHeB only).

RSCLX(I) - If positive, rescale to this hydrogen mass fraction (envelope in CHeB, full star otherwise.)

RSCLZ(I) - If positive, rescale to this metallicity (whole star).

LSENV0A(I) - If true, adjust the outer fitting point mass location.

SENV0A(I) = -1.0D-7 ! Log of fractional envelope fitting point mass, defined as the difference between the outer fitting point and the total mass. Values greater than -1e-11 are set to -1e-11, e.g. this must be a negative number. -1e-4 is standard, -1e-7 is a thin outer envelope fitting point; -1e-11 sets the fitting point at the atmospheric boundary condition, equivalent to MESA.

Optional stopping conditions:

For all of these, a negative value implies that it is not used. If the model starts below the required threshold, the condition is also disabled. These only function for an evolving run (KINDRN(I) = 1).

ENDAGE(I) - Stop runs if the model age exceeds this value (in yr).

END_DCEN(I) – Stop runs if the central deuterium drops below this level. Select a value 99% of the initial one for the start of the deuterium-burning birthline; select a value 1% of the initial one for the end of the deuterium burning birthline.

END_XCEN(I) – Stop runs when the central hydrogen drops below this level. A value 1% below the initial one is a good approximation for the zero-age main sequence; a value of 0.0001 corresponds to the terminal-age main sequence.

END_YCEN(I) - Stop runs when the central helium drops below this level. Y=1-Z-0.04 (4 percent of He burned) is a good approximation for the zero-age horizontal branch; a value of 0.0001 corresponds to the terminal-age horizontal branch. NOTE: this should only be initialized after the MS; as the central He starts low prior to core H exhaustion, it will otherwise be disabled by default.

Solar Calibration

If this option is set T, it over-writes some of the global controls. It replaces them with a series of solar calibration runs, with parameters (alpha, X, Z) determined by using partial derivatives of the prior solutions relative to the target L, R, age and Z/X (if used). The final run will have the calibrated solar values, which should then be adopted as global starting values for solar-calibrated model grids.

LCALS - Toggle to auto-calibrate solar Y and alpha to match the solar R and L at the age of the Sun. Does not enforce correct Z/X for mixture unless independently specified

LCALSOLZX – Adds an auto-calibration of Z to a target surface Z/X value.

CALSOLZX – Target surface Z/X for mixture – 0.02292 for GS98.

TOLL - Tolerance target for L in the calibration routines.

TOLR - Tolerance target for R in the calibration routines.

CALSOLAGE (4.568e9) - Target solar age (years) in the calibration routines.

Note: we now recommend that this be achieved with scripts external to the code.

Heavy Element Mixture Controls

The options below are only used if LFIRST = TRUE. It is recommended that you do this to initialize the suite of starting models that you will use and then turn these off. NOTE: These options do not work if the solar calibration is enabled.

ISETMIX (0) The CNO abundances are included in the model in addition to the total metallicity. If this option is set to 0, the abundances in the starting model are retained, whether they are consistent with the mixture used in the opacity tables or not. If it is set to 1, they will default to specific mixtures hard-coded with the AMIX string (see below). If

it is set to 2, one can implement a custom mixture (useful, for example, for alpha-enhanced models.)

AMIX ('GS98') – if ISETMIX=1, use preset CNO abundances relative to the total metallicity for different solar mixtures. Currently implemented ones are Grevesse & Sauval 1998 (GS98), Amarsi, Asplund, and Grevesse 2021 (AAG21), and the Magg et al. 2022 photospheric (M22P) and meteoritic (M22M) mixtures.

FRAC_C, FRAC_N, FRAC_O (GS98 values) – the fraction of metals in the form of C, N and O. These entries can be found in the atomic opacity tables the code uses, and they are only used for a custom mixture (ISETMIX=2).

ISETISO (0) The CNO abundances are included in the model in addition to the total metallicity. If this option is set to 0, the abundances in the starting model are retained, whether they are consistent with the mixture used in the opacity tables or not. If it is set to 1, they will default to specific mixtures hard-coded with the AMIX string (see below). If it is set to 2, one can implement a custom mixture (useful, for example, for non-solar light element abundances.)

AISO ('L21') – if ISETISO=1, use preset CNO isotope ratios and DHe3LiBeB abundances. The only currently implemented mixture is Lodders et al. 2021 (L21).

R12_13, R14_15, R16_17,R16_18 (L21 values) – Isotope ratios for C^{12}/C^{13}, N^{14}/N^{15}, O^{16}/O^{17}, and O^{16}/O^{18}. Used only if ISETISO is set to custom (2). Note – N^{15} and O^{17} are not currently used.

XH2_INI, XHE3_INI, XLI6_INI, XLI7_INI, XBE9_INI, XB10_INI, XB11_INI – Mass fractions (note – NOT NUMBER!) for light species H^2, He^3, Li^6, Li^7, Be^9, B^{10}, and B^{11}. B^{10} and B^{11} are not currently used in the code.

I/O Controls

The starting and last models include only abundances and structure variables. These options allow snapshots to be generated with more information. The contents of the output files are described in a separate document.

LSTORE - If T, store model structures at specified points to the *.store file.The default output stores only the ingredients needed to start a YREC run – header, mass, structure variables, composition, and a mixing flag.

LSTATM - If T, append an atmosphere to the *.store file. The default output contains no information for the atmosphere.

LSTENV - If T, append an envelope to the *.store file. The envelope is defined as the portion of the model between the optically thin atmosphere and the outer fitting point in

the interiors model. The default output contains no information about the envelope structure outside of the last model point.

LSTCH – If T, this enables an alternative form of the .store output that provides model structures with the interior, envelope, and atmosphere solutions stitched together for each timestep. We recommend enabling this. If this is not enabled, the atmosphere, envelope, and interior output are output with different (legacy) formats.

LSTMOD - Set the same as LSTORE.

LSTPHYS - If T, append columns with microphysics data (e.g. epsilon, kappa) to the *.store file.

LSTROT -If T, append columns with rotation data, including diffusion coefficients used for rotational mixing, to the *.store file.

NPRTMOD – Generate a snapshot of the detailed model structure every NPRTMOD models.

NPRTPT – When a snapshot is generated, output every NPRTPT spatial points (if 1000 points and NPRTPT=10, 100 lines are output).

LSTPCH - If T, output details of the final model to the *.store file. If you only want to see details of the final model, you would set NPRTMOD to -1 and LSTPCH to T.

LTRACK - If T, global properties are output to the *.track file. **This should almost always be on**.

ITRVER (0) – This is a toggle for different track file formats**. 0 (one line per model) is strongly recommended.**

LRWSH - If T, the short file retains only the details from the last model. This minimizes output file size for what can otherwise be a pretty large file, containing detailed information on intermediate model properties and numerical convergence.

LCORR - If true, output verbose details on model convergence to the *.short file. This is very useful for tracing failed runs.

LPULSE - If T, generate pulsation output. LSTATM AND LSTENV are needed to get data for those regions (required to generate useful data, but not enforced in the code, if LPULSE is T)

I/O paths.

Pointers and Physical Options (e.g. which EoS table will be used) are discussed in the namelist 2 data below. Here we only discuss the starting model and output files.

FFIRST - Starting model path.

FLAST - The last converged model is stored here.

FSTOR -Model snapshots during the run are stored here.

FTRACK - Global variables as a function of time are stored here.

FSHORT - Details about the numerics of the model run are stored here, as well as some details not currently in the track file.

FPMOD - Pulsation output for the interior model is stored here.

FPENV - Pulsation output for the envelope model is stored here.

FPATM - Pulsation output for the atmosphere model is stored here.

The following output files are depreciated but still usable:

FMODPT - No longer used.

FSNU - No longer used (formally, output if LSNU = T).

FSCOMP - No longer used (light elements vs. time, now included in track file)

Core Model Microphysics: yrec8.nml2

Standard stellar models are important tools for understanding stellar populations, and their approximations are reasonable ones for a range of interesting physical conditions. We therefore begin with ingredients in common with all models: the equation of state; opacities for conductive and radiative energy transport; convection; and the treatment of the boundary conditions. YREC is equipped to explore a number of distinct assumptions; note that some options are tied to the code being backwards-compatible with obsolete scenarios that are not currently recommended for use.

Variables with a blue prefix (YREC8.NML1) are set in the first namelist; generally speaking, these are input physics file locations used by the code, I/O controls, or run instructions. Variables in green are options that users are likely to need to specify for a run. Most of these are common physics options and input file names. Some are numerical variables with current defaults that are not recommended as optimal. Variables in black are valid options, but ones that users will typically not need to adjust. Variables in red are either experimental options, obsolete, or options that we recommend that users no longer choose.

Equations of State

The equation of state (EoS) relates composition, temperature and density to pressure. YREC employs a mixture of semi-analytic and tabulated equations of state. The EoS is also used to compute thermodynamic quantities, such as specific heats, and partial derivatives used for the solution of the structure equations (e.g. d ln P / d ln T). The latter impact the convergence of the structure equations, but do not dictate the solution; as a result, it can be important to have smoothly varying derivatives, which can be a challenge. Furthermore, different equation of state tables are generally valid only in specific domains. YREC therefore has a layered equation of state, defined as follows.

The Base YREC EoS.

The default equation of state that includes electron degeneracy pressure, radiation pressure, and a Boltzmann-Saha solver for H and He that assumes full ionization higher than a critical threshold. This EoS can also include collective effects, such as the Debye-Huckel correction for the pressure reduction due to mutual electron repulsion. It is used by default where other EoS values are not present; because it is semi-analytic, the derivatives from this EoS are used for the solution of the structure equations even when other tabulated EoS calculations are used for the pressure itself. Radiation pressure is included analytically; electron degeneracy involves a table lookup.

YREC8.NML1 FFERMI (n/a) – Path for the table of Fermi integral solutions, used for computing electron degeneracy pressure. This includes partial degeneracy and relativistic corrections. *This must be specified for the code to run.*

LDH (TRUE), **ETADH0** (-2), **ETADH1** (1) – LDH toggles the usage of the Debye-Huckel correction to the standard EoS (T/F); ETADH0 and ETADH1 are the domains where it is used. Default settings are recommended.

TSCUT (6.0) – A fully ionized EoS is assumed for log T > TSCUT. Do not alter without good reason, as the Saha equation for H and He breaks down at high density!

The OPAL EoS

The tabulated OPAL EoS (Rogers& Nayfonov 2002) is the main recommended one for stellar evolution calculations. It is available for 3.3 < log T < 8.3; there is a complex edge to the tables at higher densities, which is a function of T (see Figure 2, Rogers& Nayfonov 2002). The biggest practical restriction on the tables is the density range; there are regimes for low mass stars that are outside of the tables. If this condition is triggered, the code uses the default EoS. OPAL assumes a fixed heavy element abundance and mixture and does not allow the heavy element fraction to vary during a run (e.g. from gravitational settling). in general different OPAL tables should be used for different mixtures. Note: this is a minor effect, as P is insensitive to the metal content. I am having trouble finding the literature reference for the 2006 EoS, which I remember being a correction for an error in the treatment of relativistic electron degeneracy.

YREC8.NML1 FOPALE (n/a) – path to the OPAL EoS Tables.

LOPALE (FALSE), **LOPALE01**(FALSE), **LOPALE06**(FALSE) – These are T/F flags for the usage of the 1996, 2002 and 2006 versions of the OPAL EoS. We should alter the defaults to make LOPALE06 = TRUE and depreciate the others. LOPALE06 = TRUE is strongly recommended. If all are set, LOPAL06 takes precedence over LOPAL01, which in turn takes precedence over LOPALE.

LNUMDeriv (FALSE) – replaces the semi-analytic YREC EoS derivatives with numerical derivatives from the OPAL tables. **This choice imposes a large (factor of 3) performance penalty, with little impact on the solution. Choose wisely.**

The SCV EoS

The tabulated SCV EoS is a viable supplement to the OPAL EOS for lower temperatures. The code logic is OPAL > SCV > YREC for cases where the EoS is outside of the table domain; a ramp is employed close to the OPAL table edge boundary. SCV uses three different pure abundance tables (X, Y, and Z; the latter was actually generated using the YREC EoS assuming Boltzmann-Saha and is a minor correction), with the EoS inferred by combining the solutions in a thermodynamically consistent fashion.

YREC8.NML1 FSCVH (n/a), **FSCVHE** (n/a), **FSCVZ** (n/a) – paths to the SCV EoS tables for H, He and “metals”.

LSCV (FALSE) – A T/F flags for the usage of the SCV EoS. *Note that if the OPAL EoS is used, SCV only is adopted if outside of the OPAL tables. To get a full SCV calculation, set OPAL off and SCV on!*

The MHD EoS

The tabulated MHD EoS is not currently recommended for usage, as it has not been maintained and modernized with code improvements. It is only available for low densities in any case. FMHD1, FMHD2, FMHD3, FMHD4, FMHD5, FMHD6, FMHD7 – locations of the MHD EoS tables.

Opacities and Opacity Tables

Opacities are a crucial ingredient for stellar models; because of their complexities, they are always evaluated via a table lookup. YREC uses a local 4-point Lagrange interpolator for opacities, which performs well compared with more complex and non-local spline fits (Bahcall, Serenelli & Pinsonneault 2004). Different opacity tables are required for different domains. At low temperatures, molecules are important; we use the Ferguson et al. (2005) tables. We should add AESOPUS as an option. Atomic opacities are valid for higher temperatures, and there are two physical choices for these: OP opacities (Badnell et al. 2005) and OPAL (Iglesias & Rogers 1996). ***Both sets of tables use the same format, so the opacity source is dictated by the table that is used, not by a specific flag in the code***. We should add LANL tables as an option. For He-burning stars, the standard atomic opacity tables are not valid, and tables with a different format are required (need to look up the LANL table reference). In practice, the opacity in this domain is well-approximated by electron scattering, so results are not sensitive to this choice. Finally, conductive opacities are important at lower temperatures and higher densities (Potekhin et al. 1999).

Conductive Opacities

YREC8.NML1 FCONDOpacP (n/a) – pointer to Potekhin et al. 1999 conductive opacity tables.

LCONDOpacP (FALSE) – Set to TRUE to use the Potekhin tables; otherwise, use the older Hubbard & Lampe ones. Change default to TRUE.

Molecular Opacities

YREC8.NML1 FALEX06 (n/a) – pointer to Ferguson et al. (2005) molecular opacities. **NOTE: The adopted mixture should be checked to ensure that it is consistent with the one for the model; agreement is not checked in the code.**

YREC8.NML1 LALEX06 (FALSE), TMOLMIN (4.0**), TMOLMAX (**4.1**) –** Set to TRUE to use the 2005 tables. YREC uses these exclusively below log T = TMOLMIN and uses the atomic ones exclusively above TMOLMAX. Between the two, there is a ramp between the tables. **Setting this TRUE is strongly recommended**. I am not including

the documentation for the older versions of the table, and we should omit them going forward.

Atomic Opacities

YREC8.NML1 FLIV95 (n/a) – pointer to OPAL or OP opacities. **NOTE: The adopted mixture should be checked to ensure that it is consistent with the one for the model; agreement is not checked in the code.**

YREC8.NML1 LOPAL95 (TRUE**) –** Set to TRUE to use the 2005 tables. Oddly enough, there appears to be no default setting for this (which would make it default to false. We should default this to TRUE. YREC ramps between the atomic and molecular opacities as discussed above. It uses the PUREZ opacities when Z > 0.1. **Setting this TRUE is strongly recommended**. I am not including the documentation for the older versions of the table, and we will omit them going forward.

He-Burning Opacities

YREC8.NML1 FPUREZ (n/a) – Pointer to C/O mixture opacity tables, for usage in helium-burning regions.

YREC8.NML1 LPUREZ (FALSE) - The LLAOL pure Z tables will be used ony if LPUREZ=TRUE. Note that if this option is not used, the code will stop if Z > 0.1 (the maximum Z in the atomic opacity tables). Logical flags and options for obsolete 1989 LAOL opacities will be removed and are not discussed.

Nuclear Reaction Rates, Composition Change and Solar Neutrinos

YREC includes non-equilibrium nuclear burning of H and He, and continuum neutrino cooling. The YREC energy generation routine is fully described in Bahcall & Pinsonneault (1992) and in Bahcall, Pinsonneault & Wasserburg 1995. Gravitational potential energy, as well as rotational kinetic energy for models with rotation, are automatically considered in the energy generation rate for the structure equation solver. It uses an implicit solver for H and He burning, using local burning rates in radiative regions and a mass-weighted average in convective ones (see convection below).

There are some features for YREC that users should be aware of. YREC treats the full pp chain and CNO cycle. However, only H, ^{3}He,^{4}He, ^{12}C, ^{13}C, ^{14}N, ^{16}O, and ^{18}O are fully solved for in the non-equilibrium composition change equations. Therefore, some intermediate reactions are treated as equilibrium ones: 1) each p+p reaction is assumed to be followed immediately by p+d, producing ^{3}He. **Non-equilibrium D burning is considered only in the pre-MS (birth D is converted to ^{3}He, and when it drops below a critical threshold it is zeroed out.)** 2) Each p+14N is assumed to be followed immediately by p+15N, producing either 12C+4He or 16O depending on the branching ratio; 3) the only He-burning reactions included are triple α, α+12C, and α+14N. **The heavy element vector Z is incremented for He burning, but not for CNO processing. This is due to the assumption of a fixed heavy element mixture in the opacity tables; CNO processing does not significantly impact core opacities (Bahcall & Ulrich 1988), so incrementing Z without correcting for the mixture change produces errors in the opacities.** YREC can also compute burning of ^{6}Li, ^{7}Li, ^{9}Be, ^{10}B and ^{11}B in the trace element approximation; the code will solve for abundance changes but does not track the associated energy generation for these species.

YREC does not assume a polynomial fit between nuclear reaction rates, and the associated energy generation rates, and the local physical conditions. Instead, for H burning it computes rates from the masses and charges, under the assumption of full ionization, with user-specified cross-sections and their first and second derivatives. The default rates are from Solar Fusion II (Adelberger et al. 2011). We should update these to Solar Fusion III. **He-burning rates are strongly affected by resonances, requiring a different approach (Caughlin & Fowler 1988). The code currently uses a polynomial fit for He burning rates, not the direct calculation used for H burning.**

H and He Burning Cross-Sections and Their Derivatives.

LNEWNUC (FALSE) – Global toggle to switch to specifying absolute cross-sections, rather than ratios of the cross-sections to Bahcall & Ulirch 1988, for inputs. The reference rates are also shifted from Bahcall & Ulrich 1988 to Adelberger et al. 2011. We should make this the default.

Cross-section values. S0_A_B, S0P_A_B, S0PP_A_B are used to denote the reaction rates between A and B, their first derivatives, and their second derivatives respectively. As there are many of these (39 in total), we discuss them in a group. Specific H-burning reactions tracked are (1,1), (1,2) – in light element burning, (3,3), (3,4), (1,12), (1,13), (1,14), (1,15 to 12C), (1,15 to 16O), (1,16). Solar neutrino calculations (and energy generation) also include the pep reaction (pp mediated by an electron); hep (direct capture of a proton by 3He); (1,Be7) electron capture = ppII; (1, Be7) proton capture = pp III. Currently the He-burning reactions are not adjustable (we should modify the code to do so – a different formulation is needed for resonant rates.) They are set at the Caughlin & Fowler 1988 rates. The fast CNO cycle, important in some massive stars, is also not fully included in the current code**. Deuterium burning is currently hard-coded and treated separately from other reactions; it should be folded into the reaction network directly. Note, however, that the current abundance change solver (KEMCOM) is hard-wired only for a select set of reactions and would need to be rewritten. For reference, the NACRE II (d,p) cross section is 2.1e-4 Kev b.**

S0_1_1 (4.01e-22),**S0_3_3**(5.21e3), **S0_3_4**(5.6e-1), **S0_1_12**(1.34e0), **S0_1_13**(7.6e0), **S0_1_14**(1.66e0), **S0_1_16**(1.06e1), **S0_PEP**(3.5734e-6), **S0_1_BE7E**(1.7709e-10), **S0_1_BE7P** (2.08e-2), **S0_HEP**(8.6e-20), **S0_1_15_C12ALP**(7.3e4), **S0_1_15_O16** (3.6e1)- Cross-sections (KeV-barns), from Adelberger et al. 2011: Note that PEP is the proportionality constant of the rate relative to pp; Be7+e- is the cross section for electron capture. If using tabulated rates, YREC uses T9 = T/10^9 K, so any tabulated rate using T6 = T/10^6 K needs to be adjusted. Some cross-sections are also tabulated in MeV barns.

S0P_1_1 (4.49e-24), **S0P_3_3**(-4.9e0), **S0P_3_4**(-3.6e-4), **S0P_1_12**(2.6e-3), **S0P_1_13**(-7.83e-3), **S0P_1_14**(-3.3e-3), **S0P_1_16**(-5.4e-2), **S0P_1_BE7P** (-3.12e-5)- First derivatives of cross-sections, from Adelberger et al. 2011, in b. Note that this term is not used for all rates.

S0PP_1_12 (8.3e-5), **S0PP_1_13** (7.29e-4,), **S0PP_1_16** (0.0), **S0PP_1_BE7P** (-2.288e-7) – Second derivatives of cross sections, from Adelberger et al. 2011. Note that this term is not used for all rates.

Electron Screening and Neutrinos.

This section includes options for how to treat the reduction in the effective Coulomb barrier for nuclear reactions from the presence of free electrons (electron screening); whether solar neutrino fluxes need to be computed; and how neutrino cooling in the continuum is computed.

WEAKSCREENING (0.03) – Threshold for weak screening of nuclear reactions, using the formulation of Salpeter (1953). In later work (see Bahcall & Pinsonneault 1992), the intermediate and strong screening assumptions were questioned; **a value of 30 is now recommended, corresponding to the assumption of weak screening at all times**.

LSNU (FALSE) – Compute solar neutrino fluxes if LSNU = TRUE.

LNULOS1 (FALSE) – If this is set to TRUE (recommended), use the Itoh 1996 neutrino cooling rates in preference to the legacy YREC values.

LiBeB reaction rates.

The species 6Li, 7Li, 9Be, 10B and 11B are difficult to create and easy to destroy in stellar interiors. They are also rare, making them unimportant for nuclear reactions. YREC therefore treats them in the trace element approximation – where they are destroyed but not produced, and the associated energy generation from these rates is neglected. Note: B depletion was included in some prior code versions, but this option is not currently available; it should be added when the composition matrix is expanded.

LEXCOM (FALSE) – If set true, the light elements LiBeB are tracked in the code; this enables their destruction from nuclear reactions, rotationally induced mixing, and gravitational settling.

LXLI6 (FALSE), **LXLI7**(FALSE), **LXBE91**(FALSE), **LXBE92**(FALSE), **LXBE93**(FALSE) – These are T/F flags that allow users to input new cross sections directly (using the XS… variables), rather than the ratio of the new cross section to the default in the code (Sli…) The relevant reactions for Li are p+6Li (XLi6), p+7Li (XLi7). For 9Be, there are three branches for p+9Be, depending on product: 9Be(p,γ)10B (XBe91); 9Be(p,d)8Be (XBe92); 9Be(p,α)6Li (XBe93).

XSLI6 (5.5e3), **XSLI7**(5.2e1), **XSBE91**(1.1e0), **XSBE92**(1.5e4), **XSBE93**(1.5e4) – absolute cross-sections from Fowler et al. 1967 (except 7Li, Rolfs & Kavanaugh 1986). Current NACRE II cross-sections are 3.1e4, 5.2e1, 1.2e0, 1.5e4, 1.5e4 (we should make these the new defaults).

SLI6 (1.0), **SLI7**(1.0), **SBE91**(1.0), **SBE92**(1.0), **SBE93**(1.0) – If the logical variables are not true, these ratios relative to the defaults can be used instead. They are over-written if X values are specified.

Thresholds for Computing Energy Generation Terms.

The TCUT vector is a series of thresholds used to determine whether or not to compute nuclear reaction rates in different regimes. TCUT(1), TCUT(4), and TCUT(5) are currently used in the code. These should be re-named to more intuitive variables, and the un-used ones (2 and 3, pp and CNO respectively) should be omitted.

TCUT(1) (6.5) – Minimum log T for computing any nuclear energy generation rates and the associated equations for composition change. **This must be reduced to 5.5 if you wish to include deuterium burning in the pre-MS, and should probably have the default adjusted accordingly.**

TCUT(4) (7.7) – Minimum log T for computing He burning nuclear energy generation rates and the associated equations for composition change.

TCUT(5) (7.5) – Minimum log T for computing continuum neutrino cooling rates.

Numerical Controls for Composition Change.

YREC has a hyper-efficient implicit solver for composition change due to nuclear reactions, which requires an iterative solution. It achieves its efficiency by being customized to the reaction rates involved – e.g. it does not use a generic matrix inversion method. *This also implies that any change, such as adding new species, would require a generalization of the matrix inversion algorithm, and likely a performance hit*. Light elements (D, 6Li, 7Li, 9Be) are independent of one another, and therefore can be solved without invoking a matrix solution or iterations.

CMIN (1.0e-20) – Abundances lower than this value are zeroed out.

ABSTOL (1.0e-5) – This is the absolute error floor for the abundance solver; it is typically important for H, 4He, and He-burning products.1e-7 is recommended, and should be set as the default.

RELTOL (1.0e-4) This is the relative error floor for the abundance solver; it is typically important for H, 4He, and He-burning products. 1e-6 is recommended and should be set as the default.

KEMMAX (50) – maximum number of iterations in the abundance solver.

Convection and Overshooting

Energy transport in stars uses three channels: conduction, radiative energy transport, and convection. For conduction and radiative transport, energy is transported without bulk motions, so no mixing is induced, and the temperature gradient required to carry the flux is a function of the local physical variables. If the required temperature gradient exceeds the adiabatic one, however, the fluid becomes unstable against mass motions and convection sets in. In this case the true temperature gradient will be higher than the adiabatic one, but less than the radiative or conductive one. YREC uses the Schwarzschild / Ledoux criterion for the onset of convection and the mixing length theory, or MLT, to infer the super-adiabatic temperature gradient (Bohm-Vitense 1958). Instantaneous mixing is assumed in convection zones and nuclear burning is treated by using mass-weighted reaction rates summed across the entire convective region (see nuclear reaction rates and mixing, discussed above.)

Convective motions will penetrate into the adjacent stable layers to at least some degree. The depth of penetration is referred to in YREC as convective *overshooting*. This boundary layer phenomenon is complex to model because mixing can alter the stability conditions at the boundary, which can in turn lead to more mixing; this phenomenon is referred to as *semi-convection*. **In general, boundary layer mixing can alter both the chemical composition and the temperature gradient. In the default setting, YREC imposes mixing in overshooting regions but does not perturb the temperature gradient directly. However, to the extent that the convective criterion depends on composition, there is indirect feedback (e.g. if convection is inhibited by a mu gradient, and the region is mixed, the code is likely to find that the boundary layer is now unstable to convection; however, the code doesn't enforce an adiabatic gradient anywhere where the local stability condition is satisfied.)** There is an experimental option to consider adiabatic overshooting**. If semi-convection is being considered, the code does allow for feedback between the stability conditions and mixing. YREC also considers only "step-function" overshooting, and not other profiles.**

YREC8.NML1 CMIXLA(1-50) (n/a) – The mixing length, defined as the distance a fluid element travels in units of the pressure scale height, is a vector that is

specified in the first namelist. It is a vector rather than a scalar to allow for rescaling models from one value of mixing length to another over a series of runs. **It must be specified in all runs used in the calculation; e.g. if you have 2 runs (a rescaling run and an evolving run) you must specify CMIXLA(1) and CMIXLA(2).** CMIXLA is written to a local variable CMIXL, used in the code, and is held fixed for the course of a given run.

TGCUT (6.9) – Convection in dense regimes is very close to adiabatic; if log T > TGCUT, the mixing length calculation is skipped and Del is set to Del(ad).

LSEMIC (FALSE) – Logical T/F toggle; if set T, checks for semi-convection. This option should be enabled for core He-burning stars, and it is recommended for stars higher than 8 solar masses.

LOVSTC (FALSE) - Logical T/F toggle; if set T, enables overshooting above a central core.

ALPHAC (0.0) – extent of the core overshooting, in pressure scale heights.

LOVMAX (FALSE) - Logical T/F toggle; if set T, performs an alternate check on the extent of overshooting based on the physical size, rather than the extent in pressure scale heights. **It is important to use this if core overshooting is enabled. Formally the pressure scale height is infinite at the center, so overshoot based on a fixed fraction of a pressure scale height produces anomalous results for small cores.**

BETAC (0.15) – The core overshoot is the smaller of the target in pressure scale heights and the fraction BETAC of the physical size of the core.

LOVSTE (FALSE) - Logical T/F toggle; if set T, enables overshooting below a surface convection zone. **Note – if the fitting point is moved very far out in the star, the surface will formally be radiative; in such a case, the model will treat the "surface" convection zone as an intermediate one, and it may complicate things such as angular momentum loss from magnetized winds.**

ALPHAE (0.0) - extent of the envelope undershooting, in pressure scale heights.

LOVSTM (FALSE) - Logical T/F toggle; if set T, enables overshooting above an intermediate convection zone (and undershooting below it). **This is old, not frequently used, code, and should be carefully tested before being used.**

ALPHAM (0.0) - extent of the overshooting and undershooting for intermediate convection zones (both are set to the same value), in pressure scale heights.

LOVAD (FALSE) -Code option to make overshooting regions adiabatic. **This is very old code, whose current utility cannot be guaranteed, and it should be tested before usage.**

DPENV (1.0) – If less than one, fully mix the star from the surface to the mass fraction DPENV. **This is very old code, whose current utility cannot be guaranteed, and it should be tested before usage.**

Central and Surface Boundary Conditions

Stellar evolution is a 2-point boundary value problem, and the boundary conditions can have a strong impact on the solution. Close to the center, YREC uses a Taylor series expansion for an inner BC; this avoids divergences in logarithmic structure variables. The location of the inner fitting point is a property of the starting model rather than a model parameter.

Similarly, YREC integrates the structure equations down to a fitting point; it does not retain structure information for the outer layers. As for the central BC, this fitting point is a feature of the starting model and not directly a parameter. This differs significantly from the MESA code, which integrates from center to surface and matches the two solutions in the atmosphere. This outer layer in YREC is split into two parts – an atmosphere integration in the optically thin layers, and an envelope integration in the deeper layers.

The atmosphere solution is either a table lookup in a full model atmosphere or an integrated T-τ relation. Knowledge of L, Teff, M (or log g) and composition is required for the atmosphere solution. The bottom of the atmosphere – usually τ = 2/3 – is the initial condition for the envelope integration.

The envelope integration assumes constant L and integrates the other 3 structure equations self-consistently. The code integrates three envelope solutions close to the model L and T_{eff} to solve for the dependence of the outer BC on L and Teff, which is used in the structure equation solver. These envelope solutions are re-generated as required by the evolution of the model in the HR diagram.

The code has options to move the location of the central and outer fitting points. There is also code to generate structure variables outside of the fitting points, both for some quantities required elsewhere (such as the convective overturn timescale, or pulsation frequencies) and for i/o purposes.

Changing the Fitting Points

LCORE (FALSE), **MCORE** (0), **FCORE** (1) – If LCORE is set TRUE, the inner fitting point is moved inwards in mass by the factor FCORE; for example, if FCORE = 10, and the current inner fitting point is 0.01 of the total mass, the new inner fitting point will be at 0.001 of the total mass. The code uses a Taylor series expansion assuming constant energy generation rate and density. It assigns a number of

points based on the spatial resolution setting of the run. This approximation is close enough to permit even large changes in the fitting point to converge.

YREC8.NML1 LSENV0A(50*FALSE), **SENV0A** (50*-1.26e-4) – The outer fitting point, SENV0, is the logarithmic difference between the mass of the outer fitting point and the total mass of the star; it is therefore negative by definition. If LSENV0A is set TRUE, the code moves the outer fitting point (either inwards or outwards) to the new location SENV0A at the start of that run. It is a vector to allow users to make a series of incremental changes in the location of the boundary. This is similar to other rescaling choices, such as mass or composition, which can also be achieved with a series of rescaling steps.

Atmospheric Boundary Condition Choices

KTTAU (0) – Integer toggle for the choice of atmospheric boundary conditions. Valid choices are 0 (gray T-τ), 1 (Krishnaswami T-τ), 3 (Kurucz 1991 tabulated model atmospheres), 4 (Allard tabulated model atmospheres), 5 (Kurucz-Castelli 2004 tabulated model atmospheres.

YREC8.NML1 FATM (n/a) – Pointer to atmospheric surface boundary condition look-up tables. Note that Allard is only available for solar metallicity.

Envelope Integration Controls

TRIDT (0.01) – Spacing in log Teff between envelope integrations.

TRIDL (0.08) – Spacing in log L between envelope integrations.

For a starting model with (L,Teff) the code integrates envelopes at (L-TRIDL/2, Teff+TRIDT/2); (L-TRIDL/2, Teff-TRIDT/2); and L+TRIDL/2, Teff). *It is recommended to retain the default relative values between TRIDT and TRIDL*. Smaller triangles are more accurate if the model is changing slowly in the HR diagram. However, if the triangle size is smaller than the characteristic rate of evolution of the models, the code will extrapolate from a small baseline, leading to significantly less accurate solutions. The defaults are a compromise choice and can be altered if there are convergence issues.

TENV0 (3.0) – If the envelope Teff ever drops below this level, the code stops. **This may need to be reduced for brown dwarf models, where the surface T can drop below 1000 K.**

TENV1 (9.0) - If the envelope Teff ever rises above this level, the code stops.

In rare cases, the envelope integrator can fail, leading to a runaway to very high T or low T. This could occur, for example, if the gas pressure is very low relative to radiation pressure, triggering fail-safes in the equation of state. If this occurs, you should check the starting density of your model atmospheres, to ensure that you are not choosing a zero-point that is too radiation-dominated.

Atmosphere and Envelope Integration Numerics

The atmosphere and envelope integrators both employ an adaptive step-size, using a Runge-Kutte algorithm. They use the largest permitted step size consistent with the required error and shoot to a fixed depth in either tau (atmosphere) or P (envelope). Note that the atmospheres controls are only used if integrating an envelope within the code, as opposed to a table look-up. **If the code goes outside of the boundaries of the tabulated model atmospheres, it switches to a gray atmosphere and uses these integration controls; it is therefore important to specify them even if a table lookup is being used.**

ATMERR (3.0e-4). Target precision of the atmospheric integration (error in T of tau, evaluated by extrapolating the time step to zero). This default is much too loose; 1.0e-9 is recommended.

ATMMAX (0.5). Maximum atmospheric integration step size in tau. A smaller value of 0.25 is recommended.

ATMMIN (0.1). Minimum atmospheric integration step size in tau. A smaller value of 0.05 is recommended.

ATMBEG (0.1). Starting atmospheric integration step size in tau. A smaller value of 0.05 is recommended.

ATMD0 (1.0e-10). Starting density for the atmospheric integration.

ENVERR (3.0e-4). Target precision of the envelope integration (logarithmic error in mass, evaluated by extrapolating the time step to zero). This default is much too loose; 1.0e-9 is recommended.

ENVMAX (0.5). Maximum atmospheric integration step size in P. A smaller value of 0.25 is recommended.

ENVMIN (0.1). Minimum atmospheric integration step size in P. A smaller value of 0.05 is recommended.

ENVBEG (0.1). Starting atmospheric integration step size in P. A smaller value of 0.05 is recommended.

STOLR0 (1.0e-3). Fractional agreement between the end point of the envelope integration and the mass content of the envelope (e.g. with the default settings, the mass content of a 1.0e-4 envelope mass must agree to 1.0e-7.) This default is much too loose; 1.0e-9 is recommended.

IMAX (11) – The integrators start with a large step size and subdivide into smaller ones; they then extrapolate to zero, and the differences as the step size changes are used to infer errors. IMAX is the maximum number of loops used.

NUSE (7) NUSE is the maximum number of loops used to extrapolate to zero; if 11 levels are needed for the desired precision, for example, only the results from steps 5-11 will be used.

Dropping a Sinkline

LENVG (FALSE), **ATMSTP** (0.5), **ENVSTP** (0.5) – LENVG has been replaced by other i/o codes; omit; the ATMSTP and ENVSTP defaults are far too coarse, and should be set at 0.02. There are purposes for which it is desirable to know the structure outside of the fitting point – either for output or for cases such as inferring the convective overturn timescale for surface convection zones outside of the outer fitting point. In this case a fixed, rather than variable, step is desired; ATMSTP is the integration step in tau for the atmosphere, and ENVSTP is the integration step in P for the envelope. The integrator otherwise uses the same numerical controls as the standard atmosphere and envelope settings.

Rotation, Magnetic Fields, Mass Loss, and Gravitational Settling

The YREC code has a strong emphasis on physical processes not accounted for in classical models. In many cases, the underlying physical models are a matter of dispute; as a result, studies of these ingredients necessarily involve more options and value judgements than other physics choices. As in other aspects of the code, there are backwards-compatible options that are no longer recommended; as a result, we focus on current choices or options.

Rotation and Magnetized Winds.

- YREC includes a simplified treatment of protostar-disk coupling (see for example Denissenkov et al. 2010) and an option to initialize a uniform rotation profile for a model without rotation.
- YREC includes the structural effects of rotation, using the Kippenhahn & Thomas (1970) approach as implemented by Pinsonneault et al. (1989); these

calculations include the quadrupole component of the potential, which also impacts rotational mixing and angular momentum transport.

- YREC includes angular momentum loss from magnetized winds, with the current recommended version being the van Saders & Pinsonnneault (2013) implementation of the Matt et al. (2012) wind model. The associated dynamo model is Rossby-scaled and includes an option for saturation of the torque at high rotation rates.
- YREC includes options to vary the rotation profile enforced in convective regions, ranging from uniform rotation to constant specific angular momentum. These are assumed to be enforced instantaneously. *Convective regions are treated as single entities for mixing, angular momentum loss, and angular momentum transport calculations, both for rotation and for nuclear burning.*
- YREC include angular momentum transport, and the associated mixing, from hydrodynamic mechanics, with some additional parametric options that mimic weak and strong magnetic coupling. The diffusion coefficients are adopted from Zahn (1991), Kippenhahn (1980) and Maeder & Zahn 1998. Both angular momentum transport and mixing are treated with diffusion + advection equations under the assumption that rotation is a constant on equipotential surfaces (consistent with assumptions in the Kippenhahn & Thomas method). Key ingredients include anisotropic turbulence, such that the diffusion coefficients for mixing are reduced relative to angular momentum transport (Pinsonneault et al. 1989, Zahn & Chaboyer 1992); a solid body option, effectively reproducing the strong magnetic coupling predicted by the Taylor-Spruit magnetic instability, commonly used in other codes; a weak magnetic coupling, useful for reproducing stellar spindown and the slow rotation of the solar core (Somers & Pinsonneault 2016); and different assumptions about the inhibiting effect of mean molecular weight gradients on mixing and transport.

Rotation Namelist Options. Defaults are in (), variables that it is not recommended to adjust / use are in red.

Global

LROT (FALSE) – global T/F toggle that enables the inclusion of rotation in the code. If this is set false, the code assumes spherical symmetry.

LNEW0 (FALSE) – T/F Toggle that forces a re-calculation of the outer boundary conditions for each new model. This is automatically toggled on if LROT = TRUE (but can also be involved for models without rotation.) This is required because rotation impacts the surface boundary conditions, but the code only considers the rotation rate from the prior model in the envelope integration, so in general the SBC changes for each new model as the rotation (in general) changes.

Initialization and Star-Disk Interactions

LWNEW (FALSE), **WNEW** (0) – If set true, LWNEW initializes uniform rotation in the star at the specified angular velocity (rad/s) WNEW. (As an example, initializing at the solar rotation period of 25.4 days is WNEW = $2\pi/25.4/24/3600 = 2.863 \times 10^{-6}$ rad/s.) NOTES: This can only be done for the initial model; if it fails, you may need to do a series of runs starting with a slower rotation rate and then scaling it up.)

LDISK (FALSE), **TDISK** (0), **PDISK** (0) – if LDISK is true, this enforces co-rotation with an accretion disk at a specified rotation period in rad/s (PDISK) up until an age of TDISK (in Gyr). Note that this is related to the absolute model age and not to the elapsed time in a given run.

Angular Momentum Loss

LJDOT0 (FALSE)– If set true, angular momentum loss is included in the model if there is a surface convection zone. Note that this can result in no applied torque if the outer fitting point is below the depth of a thin surface convection zone! (NOTE: the outer point is assumed to include all of the mass between it and the surface for angular momentum purposes.)

There are a number of parameters tied to the legacy wind model; it is strongly recommended to adopt the new formulation, that includes the old one as a special case.

LMWIND (TRUE) – use the Matt et al. (2012) wind formulism, which can be expressed as

$$\frac{dJ}{dt} = \frac{\omega K_1^2}{(K_2^2 v_{esc}^2 + \omega^2 R^2)} B^{4PMMM} \left(\frac{dm}{dt}\right)^{1-2PMMM} R^{2+5PMMM} M^{-PMMM}$$

The B and dm/dt values are adopted from a scaled solar model, where

$$\left(\frac{B}{B_{sun}}\right) / \left(\frac{R}{R_{sun}}\right)^{PMMC} = \left(\frac{\omega\, \tau_{cz}}{\omega_{sun} \tau_{cz,sun}}\right)^{PMMB} \left(\frac{P_{phot}}{P_{phot,sun}}\right)^{0.5}$$

and

$$\left(\frac{dm/dt}{dm/dt_{sun}}\right) \propto \left(\frac{\omega\, \tau_{cz}}{\omega_{sun} \tau_{cz,sun}}\right)^{PMMA}$$

where the product $\omega\ \tau_{cz}$ is capped at a saturation threshold $\omega_{sat,sun}\ \tau_{cz,sun}$.

AWIND (V13) – There are three currently implemented options. K97 uses the legacy Krishnamurthi et al. 1997 wind model (Kawaler 1988, with a Rossby scaled saturation threshold; PMMM=0.5,PMMA=2,PMMB=1,PMMC=2); V13 uses the van Saders & Pinsonneault (2013) wind mode (PMMM=0.22,PMMA=2,PMMB=1,PMMC=0); and CUS (for custom) allows users to manually specify the free parameters in the Matt et al. (2012) wind model.

FK (1.0) – Scale factor for the magnetized wind torque, which is scaled relative to the solar value. This is the equivalent of adjusting some combination of the solar coronal magnetic field strength and mass loss rate.

LROSSBY (FALSE) – If TRUE (recommended), the field strength is scaled by the ratio of the stellar convective overturn timescale to the solar one; to compensate, the saturation threshold is also adjusted.

WMAX (3e-4), **WMAX_SUN** (1000). The wind model assumes that the torque scales as ω^3 (for the V13 and K97 models). This can either be capped at an absolute ω (WMAX) or one relative to the solar ω (WMAX_SUN). The code defaults to the latter if it is less than the default value. These values scale with the Rossby number if LROSSBY is TRUE; typical values for spindown models are of order 10 times solar.

PMMJD (1.32e30), **PMMMD** (1.27e12), **PMMSOLP** (4.93e4), **PMMSOLW** (2.836e-6), **PMMSOLTAU** (9.4805e5). The magnetized wind is a scaled version of the solar wind; these entries specify the zero-points used in the calculation for the solar wind torque (cgs), mass loss rate (g/s), log of the pressure at τ=2/3, used to compute the equipartition atmospheric field strength; angular velocity (rad/s), and overturn timescale (s). The pressure and overturn timescale can reasonably be varied with a calibrated solar model; the other three (in red) should generally be held fixed, with the zero point of the torque adjusted with the FK variable (above).

PMMA (2), **PMMB**(1), **PMMC**(2), **PMMD**(0), **PMMM** (0.5) – These are individually specified and used only if AWIND=CUS. Note that PMMD is needed to adjust for a slight difference in functional form between the K97 and Matt wind laws, and PMMC should be set to zero unless mimicking the K97 loss law; neither is not used in general otherwise. These could be adjusted to explore, for example, deviations from the Skumanich spin down law exponent. The default should be switched to V13 settings.

ALFA (1.5) – Legacy wind index, not used if LMWIND=TRUE.

The Structural Effects of Rotation

YREC includes the structural effects of rotation, assuming constant rotation on equipotential surfaces. The axisymmetric shape of the equipotential surfaces is computed following the Kippenhahn & Thomas (1970) method, which included the spherically and cylindrically symmetric components of the potential. YREC also includes the contribution from the quadrupole moment (added in Pinsonneault et al. 1989). The code stores a shape term ETA2 and an averaged radius R0 (the latter is the radius that the distorted surface would have if it was a sphere; the former assumes that the shape can be modeled as R0 (1- a*P2(cos θ)). The code also solves for correction terms FP and FT to the structure equations. There are a few options related to rotation:

ACFPFT (1e-36), **ITFP1** (5), **ITFP2** (20) – These are related to the precision of the solution for the shape of the equipotentials. The first is the cube of the precision of the shape term; the second and third are the number of iterations permitted to solve for R0 and ETA2. Unless rotation is approaching break-up, these should be more than enough. **Note that YREC will halt if rotation exceeds break-up anywhere in the star, and it**

is likely that models above 90% of critical will be numerically touchy. Formally, however, the code is precise and accurate to 99% of surface breakup.

ITDIF2 (1) – number of times that the code iterates between the diffusion coefficient calculation and mixing, which captures the impact of rotationally induced mixing on the mixing rate. For models with rotationally induced mixing, a value between 2 and 10 is recommended.

Angular Momentum Transport and Rotationally Induce Mixing

Global Options.

LINSTB (FALSE)– LINSTB is a global T/F toggle for computing angular momentum transport and the associated mixing from hydrodynamic instabilities in radiative regions. If set FALSE, there is no rotationally induced mixing. Angular momentum is then conserved locally, unless some other global option, like solid body rotation, is enforced (see below).

LSOLID (FALSE) – LSOLID is a T/F toggle. If set TRUE, it enforces solid body rotation at all times in the entire star; permits rotationally induced mixing. This toggle produces very close to those involving the Spruit-Taylor magnetic instability in the absence of strong mean molecular weight gradients and can be though of as a strong magnetic coupling case.

LNOJ (FALSE) – LNOJ is a T/F toggle. If set TRUE, it allows mixing, but not angular momentum transport, in radiative regions; this is a specialized case that can be invoked to investigate mixing with local angular momentum conservation (if LSOLID = TRUE it is over-written).

WALPCZ (0). Global toggle enforcing a rotation profile $\omega \sim R^{WALPCZ}$; zero corresponds to solid body rotation and is the default. Note that if WALPCZ is not zero, there is no requirement imposed on the adjacent radiative zone unless IMPJMOD (below) is used. WALPCZ is bounded to be between 0 and -2, the latter corresponding to uniform specific angular momentum.

IMPJMOD (0) –0 is no code change; 1 is solid body rotation in all cases, including the CZ (over-writes WALPCZ, equivalent to LSOLID); .2 allows the envelope to rotation differentially, and have the core rotate uniformly (equivalent to LSOLID=T and WALPCZ not zero); 3 allows the code to couple differential rotation in the outer convection zone with solid body rotation in the interior, with a value equal to the base of the CZ. (JT code extension, not available in some versions). **Note that 2 and 3 are new scenarios not possible in prior versions of the code (solid body core / differential envelope).**

Angular momentum transport coefficients.

The code computes and sums diffusion coefficients for angular momentum transport hat include meridional circulation, the secular shear, the GSF instability, and a diffusive

magnetic term. The meridional circulation coefficient is derived from Zahn (1991), retaining only terms of order the first derivative in angular velocity; GSF is from KIppenhahn (1980); secular shear is from Zahn 1991 as well. Note that the code also checks for dynamical instabilities (dynamical shear and Solberg-Hoiland), which are used to set limits on the maximum permitted shears. In practice these are rarely triggered.

FW (1), **FES** (1), **FGSF** (1), **FSS** (1) – FW is a global multiplicative factor for all hydrodynamic diffusion coefficients; while FES, FGSF, and FSS multiply only the meridional circulation, Goldreich-Schubert-Frick (GSF)+ABCD, and Secular Shear (SS) ones. While useful for parameter variations, these should not otherwise be altered without good reason.

FC (1), **FESC** (1), **FGSFC** (1), **FSSC** (1) – **FC is the scale factor between coefficients for mixing and angular momentum transport; it is essential to calibrate this for rotationally induced mixing, typically using Li depletion** (Somers & Pinsonneault 2016). FESC, FGSFC, and FSSC are related scale factors for only the meridional circulation, Goldreich-Schubert-Frick (GSF)+ABCD, and Secular Shear (SS) ones. While useful for parameter variations, these should not be altered without good reason.

LVFC (FALSE), **LDIFAD** (FALSE). Option to use the Zahn (1991) formula relating the angular momentum and mixing diffusion coefficients, treating FC as a function of position in the star. This is experimental, and overwrites FC, FESC, FGSFC, and FSSC. FDIFAD includes both the diffusion and advection terms in meridional circulation from Zahn 1991. This is a reasonable physical choice to invoke (see Maeder & Zahn 1998). **LDIFAD is currently not working properly and should be disabled until fixed.**

LCODM (FALSE), **CODM** (2.5e4). Option to enable a term that transports angular momentum without mixing; this approximates a diffusive magnetic term (Somers & Pinsonneault 2016). This term can be mapped onto a timescale for core-envelope decoupling (Denissenkov et al. 2010).

The sensitivity of meridional circulation to mu gradients is taken from Zahn & Maeder (1998). There are additional physical options for angular momentum transport below.

IES (1), **IMU** (1), **FO** (1) – no longer used.

IGSF (1). There are three possible stability conditions for the GSF instability. 0 inhibits it by viscosity and mu gradients; 1 includes neither; 2 includes viscosity only. 1 is the original Kippenhahn estimate (and is appropriate if the ABCD instability is included).

RCRIT (1e3) – Threshold for triggering the secular shear instability; treat as fixed.

Gravitational Settling.

YREC includes gravitational settling and thermal diffusion of helium and heavy elements with respect to hydrogen, and includes different physical scenarios for studying them. Our adopted formulism is that of Bahcall & Loeb (1989), with diffusion coefficients computed using the Thoul (1994) approach, which in turn is based on the Burgers physical model. **The code assumes that all heavy species diffuse at the same rate as fully ionized iron, even though the code can compute element-specific diffusion rates. This is because the adopted input physics (model atmospheres, equation of state, and opacity tables) assume a fixed heavy element mixture. Deviations for solar models in this approximation are small, of order 1% (Richard et al. 1996); however, in chemically peculiar stars this need not be true. Light elements (LiBeB) do not impact the microphysics and diffuse at element specific rates, as does He.** The current code therefore also does not include radiative acceleration (but modifications to do so would be straightforward.) **Note also that the current code solves for gravitational settling, rotational mixing, and nuclear burning in separate routines, not unified into one.**

It is currently recommended to use the 2015 diffusion subroutines, which supersede the prior ones; this overwrites many of the other code options or renders them irrelevant, except for backwards compatibility. This toggle must be enabled to compute gravitational settling of light species.

LNEWDIF (FALSE) , **LDIFLI** (FALSE) , **LDIFY** (FALSE), **LDIFZ** (FALSE)– LNEWDIF enables the new diffusion routines, which include the diffusion of light species. We will be switching to setting LNEWDIF TRUE by default. LDIFLI enables the diffusion of light species; LDIFY does so for helium; and LDIFZ does so for metals. In general, it is recommended to set all three TRUE, or all three FALSE, for models with and without diffusion.

FGRY (1), **FGRZ** (1), - scale factors multiplying the diffusion coefficients for helium and heavy element diffusion respectively. Thoul (1994) estimated uncertainties are at the 15% level; this can also be reduced from unity to mimic the effects of rotationally induced mixing in models without rotation.

DT_GS (0.1)**, XMIN** (0.001)**, YMIN** (0.001)**, GRTOL** (1e-8)**, NITER_GS** (10) – Numerical controls for the gravitational settling abundance solver. DT_GS is the maximum timestep relative to the gravitational settling timescale at the base of the surface CZ; if the timestep is too large, the code breaks it up into a series of smaller ones. The code shuts down diffusion if the central hydrogen drops below XMIN the diffusion calculations are shut off (this is tied to convergence issues in the Bahcall & Loeb formulation); if the helium abundance is everywhere below YMIN diffusion is also turned off (this would be quite pathological). GRTOL is the convergence criterion for the abundance changes, and NITER_GS is the maximum number of iterations allowed to reach it. There is ordinarily no reason to adjust these.

ILAMBDA (1), **LTHOULFIT (FALSE)**, **LTHOUL (FALSE) –** We need to adjust the code to set all of these toggles correctly for the new diffusion routine; at present this is done only if LDIFLI = T. Note that none of these are used for the new routine (so perhaps just adjust the defaults for them.) ILAMBDA allows different choices for the treatment of the Coulomb logarithm, used for the collision integrals; LTHOULFIT is used for an analytic fit to the full Thoul coefficients, which is not recommended; and LTHOUL uses a more general routine for diffusion coefficients, as opposed to the default (hard wired for fully ionized He and Fe). If NEWDIF is TRUE, any values entered are over-written.

Star Spots and Magnetic Fields.

YREC includes the suppression of convective flux in starspots, and the associated change in the atmospheric surface boundary conditions**.** For a detailed description of the model and associated isochrones, see Somers, Cao & Pinsonneault 2020. The current code supports only a constant spot filling factor; experimental code will couple this to the Rossby number and a saturation threshold.

SPOTF (0), **SPOTX** (1), **LSDEPTH** (FALSE) – SPOTF is the fraction of the surface area covered by star spots, which have a temperature contrast relative to the ambient photosphere of SPOTX. 0.85 is a typical value for sunspots and recommended if they are being used. LSDEPTH is a T/F toggle that determines whether the temperature contrast is modeled as a constant, or if it is allowed to decrease with depth. The latter case assumes a fixed magnetic field strength with depth, which will increase SPOTX as the ambient pressure rises.

Mass Loss.

YREC has experimental options to include mass loss; at present, these are not self-consistently implemented to include the effects of mass loss on rotation. The mass loss from magnetized winds is also not, at present, included in models.

LSOLWIND (FALSE) – Currently a placeholder for including the mass loss from solar-type winds; as the mass loss rate is included in the angular momentum loss rate, this option will require LJDOT0 and LMWIND = TRUE.

LMDOT (FALSE), **DMDT0** (-1e-14), **LREIMER** (FALSE), **CREIM**(0). LMDOT is a global toggle for including mass loss (T/F). DMDT0 is a constant mass loss rate; expressed in solar masses per year. LREIMER is a T/F flag that enables a mass loss rate using the Reimers' formula, with a scaling constant CREIM (equivalent to eta). Note that there is experimental code to enable mass accretion, with a positive DMDT0, and an associated abundance mixture that need not be the same as the underlying model. Test before using!

YREC Numerical Controls

YREC solves for the evolution of a stellar model using a finite difference method. This involves replacing continuous functions in space and time with discrete intervals. The correct choices are necessarily a matter of judgement; finer resolution will result in more precise, but slower, models. **The choice of spatial and temporal resolution is therefore an important consideration for the precision and accuracy of the models.** YREC also uses a relaxation method, where a prior correct solution of the structure equations is used as an initial guess for the new solution. The difference between analytic and numerical derivatives is used to infer corrections to the structure in an iterative process. **YREC defines a model as converged when the corrections drop below a user-specified threshold. It halts a run when the corrections grow too large, or when the model has taken too many iterations to converge.** In general, stellar structure codes can be tuned to be very precise for a given assumed spatial and temporal resolution. However, in some cases, interpolation errors in tables, or choices related to the treatment of mixing, can set a high error floor. **A common failure mode in YREC is that the corrections to the structure oscillate at a level higher than the target precision. Reducing the target precision, or increasing the simulation resolution, can resolve these problems.** Finally, there are some constants, and some choices related to the speed of convergence, that need to be specified.

Physical Constants

YREC does calculations in cgs but reports them in solar units. The solar L and R are therefore variables in the code. **The defaults are old; use the values below**.

CLSUN (3.8515e33) - Definition of Lsun (erg s-1) used for tracks and output. The current recommended value is 3.828e33.

CRSUN (6.9598e10) - Definition of Rsun (cm) used for tracks and output. The current recommended value is 6.957e10

Controls to Adjust Initial Mixture

These will be depreciated in favor of a more general scheme for specifying the heavy element mixture, in yrec8.nml1.**LNEWCP** (FALSE)- T/F flag to manually adjust one input abundance vector to a new uniform value.

ANEWCP (H2 ') - ID of species to adjust.

ATMP ('ABS') - Indicate if abundance is absolute or relative to H

XNEWCP (n/a) - New mass fraction; `1.5e-5 for D2.

Variable Naming Control

LNEWVARS (FALSE) – This is a global T/F flag, replacing older, very unintuitive, variable names with more useful ones. Setting this TRUE Is strongly recommended.

The code uses the new variables to generate the correct values for the old ones, which are still used in the code itself.

Controls for Setting the Number of Model Points

The following set of variables set the spatial tolerances for allocating mesh points; they replace the HPTTOL vector that the code used previously. **An important note: the current code uses the spacings here to generate a running index, allocating new points precisely where indicated. Users therefore have complete control over the spatial resolution.** The only exception is that sharp edges are preserved where real discontinuities exist (such as the edge of convective cores).

In general, the mass spacings are the most stringent in the deep core. Luminosity changes set the resolution in nuclear burning regions. Pressure changes set the resolution in the outer layers. Users should set their spatial resolution by changing them, and seeing that their results for the properties of interest vary by less than the amount that they need for their problem.

TOL_DM_MIN (1.0e-8). This is the minimum spacing in mass between points. This typically does not need to be adjusted unless the user is including a very thin envelope with a fitting point far out in the star.

TOL_DM_MAX (0.08) - The maximum spacing in log M between points; important for stellar cores.

TOL_DL_MAX (0.02) – The maximum spacing in L/Lsurf; this is important for energy generating regions, and this parameter is particularly important for the accuracy and precision of ages.

For pressure, there are different spacings possible in the radiative core, the convective envelope, and the boundary region close to the base of the surface CZ.

TOL_DP_ENV_MAX (0.05) - Maximum spacing in log P, surface CZ (most important for outer layers)

TOL_DP_CZBASE_MAX (0.05) - Maximum spacing in log P near the SCZ base

TOL_DP_CORE_MAX (0.05) - Maximum spacing in log P, interior to SCZ (important between burning, SCZ base)

TOL_CZBASE_FINE_WIDTH (0.0) - Width of the finely zoned (in pressure) region around the CZ base.

FLAG_DX (0.05) - Discontinuities in X larger than this are preserved / flagged; points at these locations are not removed during the point allocation process.

FLAG_DZ (1.0) - Discontinuities in Z larger than this are preserved / flagged; points at these locations are not removed during the point allocation process.

FLAG_DW (0.1) – For models with rotation only. Discontinuities in Omega(rotation) larger than this are preserved / flagged; points at these locations are not removed during the point allocation process.

TOL_DX_MAX (1.0) - Maximum spacing in X (obsolete, preserved for back compatibility)

TOL_DZ_MAX (1.0) - Maximum spacing in Z (obsolete, preserved for back compatibility)

Timestep Controls

The following variables set the timestep used in the model; they replace ATIME, DTWIND, and DTDIF. YREC. YREC computes all relevant ones for the model and uses the shortest time step. **There are important considerations for pre-MS models below.**

TIME_CORE_MIN (0.002) - Minimum X, Y for using "core" vs. "shell" burning criteria

TIME_DX_CORE_TOT (0.02) - Maximum absolute delta X H core burning (main sequence). This controls the timestep for most of the main sequence. Default should probably be smaller, 0.005; there are few MS models in any case and it does impact age precision.

TIME_DX_CORE_FRAC (0.5) - Maximum fractional delta X, H core burning (main sequence). Default is too large, 0.1 would be better. Controls the timestep at H core exhaustion.

TIME_DY_CORE_TOT (0.02) - Maximum absolute delta Y/Z per timestep (He core burning) (red clump/HB).

TIME_DY_CORE_FRAC (0.5) - Maximum fractional delta Y/Z per timestep (He core burning) (red clump/HB).

TIME_DX_TOTAL (0.0015) - Maximum absolute H burned per timestep, Msun (post-MS). This is a global H burning time step limit, relevant for the RGB and later.

TIME_DX_SHELL (0.1) - Maximum fractional delta X, H shell (post-MS). This is a local H shell criterion, relevant for the RGB and later.

TIME_DY_TOTAL (0.0015) -! Maximum absolute He burned per timestep in a shell source, Msun (AGB).

TIME_DY_SHELL (0.1) - Maximum fractional He burned in a shell source (AGB.)

LSTRUCT_TIME (FALSE) - Flag to restrict timestep from structural changes. **Note this is hard-coded to disable once stars reach the MS, and will not work for more evolved stars. This must be turned on to run pre-MS models. This really should default F for most cases and T should be used for the pre-MS.**

LTRIST (FALSE)- Flag to restrict timestep such that changes in L and T are smaller than the envelope triangle (specified by TRIDL and TRIDT). Only enabled when LSTRUCT_TIME is disabled (MS and onward).

TIME_DT (0.02) - Maximum local change in log T (pre-MS)

TIME_DP (0.04) - Maximum local change in log P (pre-MS)

TIME_DR (0.02) - Maximum local change in log R (pre-MS)

TIME_DL (0.02) - Maximum local fractional change in luminosity (pre-MS)

TIME_MAX_DT_FRAC (1.5) -Maximum fractional change in timestep, model to model. Regardless of other settings, the timestep cannot change by more than this ratio (larger or smaller).

TIME_DW_MIX (0.08) - Only for models with rotation. This is the maximum fractional change in angular velocity per timestep within a model. If this is less than time_dw_global, rotational evolution will be solved for in a series of smaller timesteps for each model. (e.g. if this is set at 0.01, and time_dw_global is set to 0.08, the code will take up to 8 smaller timesteps to resolve rotational mixing and angular momentum transport for each model.)

TIME_DW_GLOBAL (0.08) -Only for models with rotation. Maximum fractional change in angular velocity.

Convergence Criteria

YREC solves for structure corrections between the start and end of the timestep, and it does so iteratively. The first set of controls involve how the starting model is initialized, the number of iterations allowed before the model fails, and how aggressively corrections are applied.

LNEWS (FALSE) – If set TRUE, YREC extrapolates the changes between model (n-1) and model (n) to predict the expected final state for model (n+1). If set TRUE (recommended), this usually speeds convergence.

Iterations used: **Niter1 and Niter2 are mandatory; Niter3 and Niter4 can be nonzero for high precision work, but should be set to zero if you are having convergence issues.**

NITER1 (2) – YREC uses the prior surface boundary conditions in iteration level 1. It is recommended to do this for a couple of steps before re-evaluating them.

NITER2 (20) - The surface boundary conditions are checked at the start of the second level of iterations. This level uses the start of timestep conditions to infer energy generation rates and the rate of composition change. **20-40 steps is recommended; note that iteration levels 3 and 4 are optional.**

NITER3 (2) – The nuclear reaction rates and composition change equations use the end of timestep structure from the second level of iteration, rather than the start of timestep values. If used, a value of 20-40 is recommended.

NITER4 (0) This repeats the third level of iteration, but also is the number of times that the code iterates between structure and rotation solutions; in effect, this allows the code to include feedback from changes in the structural effects of rotation and rotationally induced mixing on the solution of the structure equations. **NITER4 can be set to zero for models without rotation; a non-zero setting simply repeats the 3rd level, representing a refinement on the solution.** If used, a value of 20-40 is recommended.

ITDIF1 (1) – number of times that the code iterates between the structure corrections and the rotation solution; only enabled if NITER4 > 0. Because the structural effects of rotation are modest in most cases, no more than 2 should be required unless very rapid rotation is being considered.

FCORR0 (0.6) - The code predicts corrections to the structure variables, which can initially be overestimated. This is the iteration 1 scale factor by which the trial corrections predicted by the code are modified.

FCORRI (0.05) - This is the increment to the FCORR0 scale factor for each successive interation, for example it reaches the max of 1 in 8 steps if (0.6,0.05). In general, a larger factor produces faster convergence, but is potentially less stable.

Convergence Criteria

The code finds the largest fractional change anywhere in the model for the 4 state variables If these maximum corrections are lower than the minimum ones below (for all 4 criteria), the run succeeds; if they exceed any of the maximum thresholds below, the run fails. Note that HTOLER (5,1) and (5,2) are applied to the right hand side of the difference equations; HTOLER (1,.) – (4, .) are applied to the left hand side. Only one set needs to be satisfied for a converged model. **Convergence criteria 100x more stringent are recommended for ordinary use; 1e4 better for solar models. The values below should be significantly loosened only if the code is having difficulty finding a solution, and ideally only for a portion of a evolutionary track.**

HTOLER(1,1) (6.0e-5) - Convergence criterion for the change in log P.

HTOLER(2,1) (4.5e-5) - Convergence criterion for the change in log T.

HTOLER(3,1) (3.0e-5) -! Convergence criterion for the change in log R.

HTOLER(4,1) (9.0e-5) - Convergence criterion for the change in L/Ltot.

HTOLER(5,1) (3.0e-5) - If the difference between analytic and numerical derivatives is smaller than this everywhere, iteration is skipped. (5,1) is for P+R

HTOLER(5,2) (2.5e-6) - If the difference between analytic and numerical derivatives is better than this, iteration is skipped. (5,2) is for T+L

HTOLER(1,2) (0.9) – If the maximum local correction in log P larger than this it ends the run. Some other bad thing usually happens first.

HTOLER(2,2) (0.5) – If the maximum local correction in log T larger than this it ends the run. Some other bad thing usually happens first.

HTOLER(3,2) (0.5) – If the maximum local correction in log R larger than this it ends the run. Some other bad thing usually happens first.

HTOLER(4,2) (2.0) – If the maximum local correction in L/Ltot larger than this it ends the run. This is usually the end state for the He flash.